

Amazon Kinesis Data Stream Hands-On Lab

To Begin with the Lab

Summary of the Lab

In this lab, **Amazon Kinesis Data Stream** was created and tested practically using a producer application and the Kinesis data viewer. The stream was created in **Asia Pacific Mumbai** with **On-Demand** capacity so AWS could adjust throughput automatically instead of manually configuring shards. The video explained that shards control stream capacity, similar to lanes on a highway, and that Kinesis stores records temporarily based on the retention period before a consumer reads them. To test the stream, a producer app was generated through an AWS CloudFormation-based setup using a Cognito user, then it was used to send sample records into the stream. Finally, the data viewer was used to confirm that records were received successfully by the stream.

Understanding Kinesis Data Stream

Amazon Kinesis Data Streams is used for real-time data ingestion where a producer sends data and a consumer reads it later. It exists so applications can continuously push streaming data without needing direct one-to-one communication between producer and consumer. In this lab, the stream acts as the buffer between the producer app and future consumers. The key concepts are **capacity mode**, **shards**, and **retention period**—with shards controlling throughput and retention controlling how long the stream keeps records available for reading.

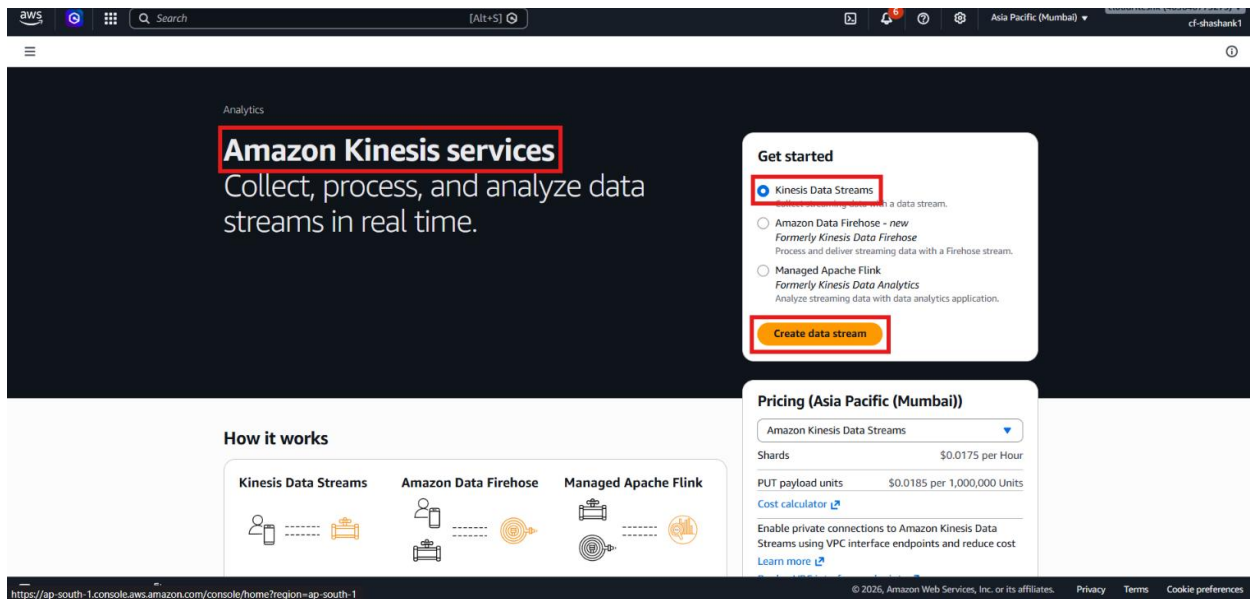
Example:

A producer app sends text such as “Hello, how are you?” into a stream named **CloudFreeks**. Kinesis stores that record for a temporary period, and a consumer can later read it from the stream. If traffic increases, shard capacity becomes important in provisioned mode, while on-demand mode lets AWS adjust automatically.

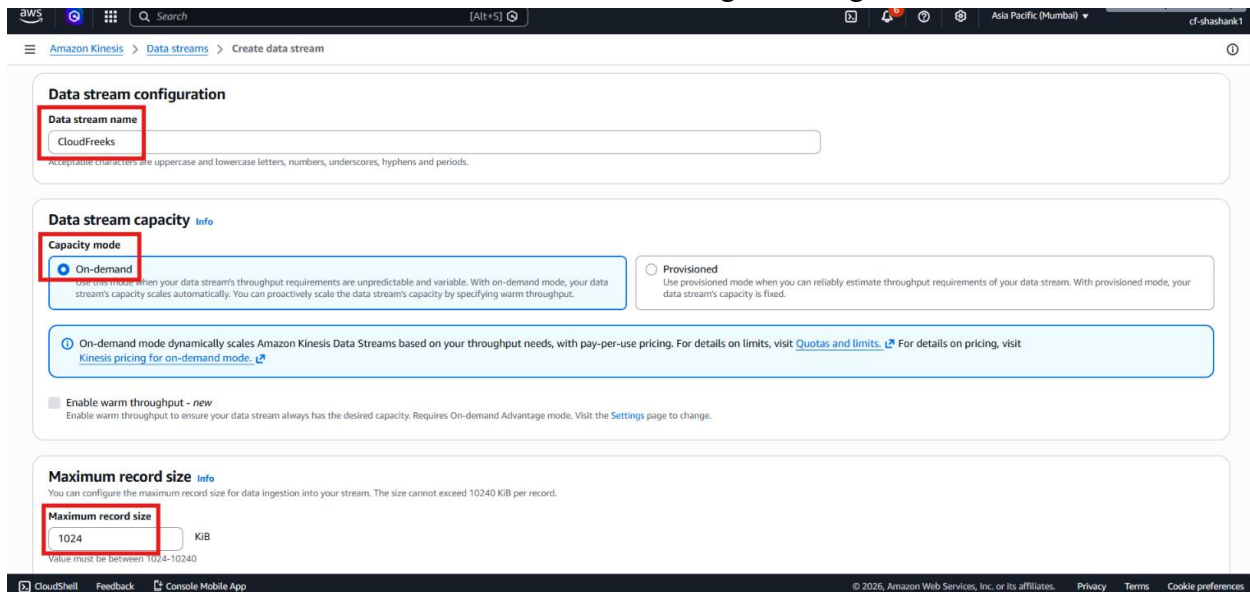
Lab Steps

Step 1 – Create the Kinesis Data Stream

- Sign in to the AWS Management Console.
- Open **Amazon Kinesis**.
- Select **Kinesis Data Stream**.
- Click **Create data stream**.



- Enter the stream name: CloudFrees
- Choose **On-demand** capacity mode for the lab.
- Leave maximum record size and other default settings unchanged.



- Create the stream in **Asia Pacific Mumbai**.

Step 2 – Understand the Stream Settings

- Review the difference between **On-demand** and **Provisioned** capacity.
- Understand that shards determine stream throughput.
- Review the retention period.
- Note that Kinesis stores records temporarily before the consumer reads them.
- Keep the default settings for the practical test.

Step 3 – Create the Producer Application

- Open the AWS-provided producer app link from the Resources and open it or directly from here.
- <https://awslabs.github.io/amazon-kinesis-data-generator/web/help.html>
- Click **Create Cognito user with CloudFormation**.

Amazon Kinesis Data Generator

- An IAM role that is assigned to authenticated Cognito users. This role has only enough permission to use the KDG.
- An IAM role that is assigned to unauthenticated Cognito users. This role has only enough permission to create Cognito analytics events.
- A Lambda function to bootstrap the Cognito Lambda install from GitHub
- A Lambda function to set up Cognito.

The Cognito Lambda function will create the following resources in your AWS account:

- A Cognito User Pool.
- A Cognito Federated Identity Pool.
- A Cognito User, with the username and password specified by you when you created the CloudFormation stack.
- The necessary relationships between the roles, users and pools.

Create the CloudFormation stack by clicking the button below. It will take you to the AWS CloudFormation console and start the stack creation wizard. You only need to provide a Username and Password for the user that you will use to log in to the KDG. Accept the defaults for any other options presented by CloudFormation.

Please note that the CloudFormation Template below is currently supported in the following regions:

- ap-northeast-1 (Tokyo)
- ap-northeast-2 (Seoul)
- ap-south-1 (Mumbai)
- ap-southeast-1 (Singapore)
- ap-southeast-2 (Sydney)
- ca-central-1 (Canada Central)
- eu-west-1 (Ireland)
- eu-west-2 (London)
- eu-central-1 (Frankfurt)
- us-east-1 (N. Virginia)
- us-east-2 (Ohio)
- us-west-2 (Oregon)

Create a Cognito User with CloudFormation

In addition to the above regions, **GovCloud** is also supported by manually importing the CloudFormation template in the following regions:

- Sign in to AWS if prompted.
- Choose the correct region for your stream.
- Create the CloudFormation stack.

Step 1: Create stack

Step 2: Specify stack details

Step 3: Configure stack options

Step 4: Review and create

Create stack

Prerequisite - Prepare template

You can also create a template by scanning your existing resources in the [IaC generator](#).

Prepare template

Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.

☒ Choose an existing template

☐ Build from Infrastructure Composer

Specify template

This [GitHub repository](#) contains sample CloudFormation templates that can help you get started on new infrastructure projects. [Learn more](#)

Template source

Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.

☒ Amazon S3 URL

Provide an Amazon S3 URL to your template.

☐ Upload a template file

Upload your template directly to the console.

☐ Sync from Git

Sync a template from your Git repository.

Amazon S3 URL

<https://aws-kdg-tools.s3.us-west-2.amazonaws.com/cognito-setup.yaml>

Amazon S3 template URL

S3 URL: <https://aws-kdg-tools.s3.us-west-2.amazonaws.com/cognito-setup.yaml>

[View in Infrastructure Composer](#)

[Cancel](#) [Next](#)

- Enter a username and password for the producer app login.
- And password should not contain special character, only Uppercase, lowercase and numbers are allowed.

Provide a stack name

Stack name
Kinesis-Data-Generator-Cognito-User
Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 35/128.

Parameters
Parameters are defined in your template and allow you to input custom values when you create or update a stack.

Cognito User for Kinesis Data Generator

Username
The username of the user you want to create in Amazon Cognito.
Shashank

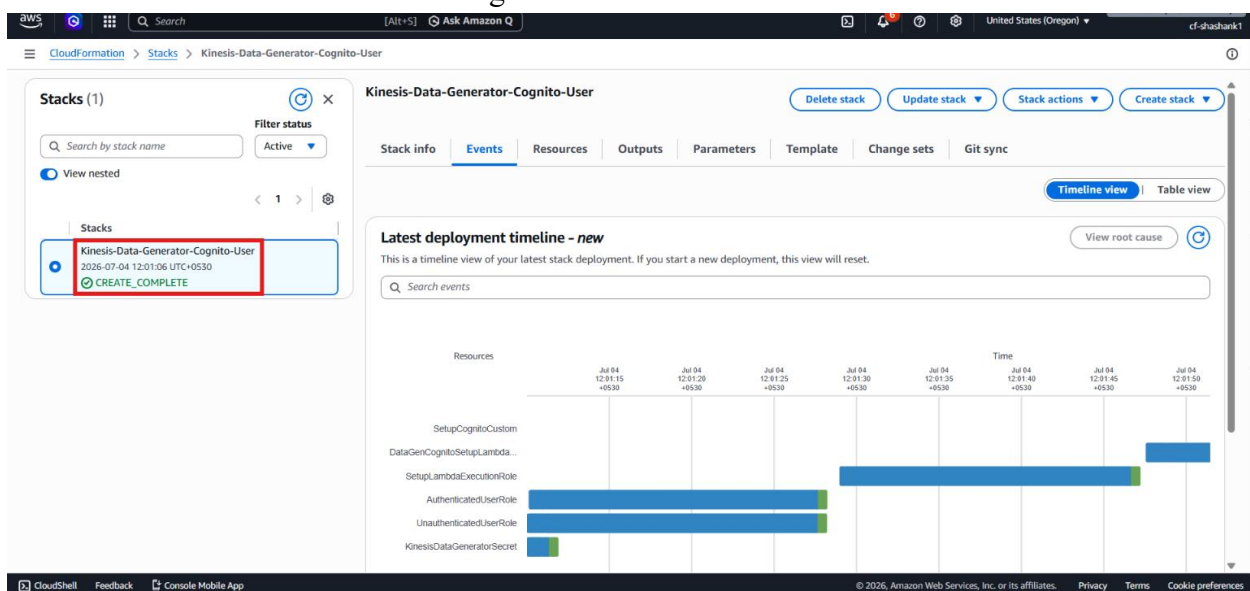
Password
The password of the user you want to create in Amazon Cognito.

Optional / Advanced Parameters (OK to ignore)

PermissionsBoundaryArn
OPTIONAL - IAM Permissions Boundary Policy ARN to attach to new IAM Roles
Enter String

Cancel Previous **Next**

- Submit the stack creation.
- Wait for the stack to finish creating.



Step 4 – Open the Producer App

- Open the **Stack Outputs** link.

Amazon Kinesis Data Generator

Configure Help Log Out

Region: ap-south-1

Stream/delivery stream: CloudFreeks

Records per second: Constant (Periodic)

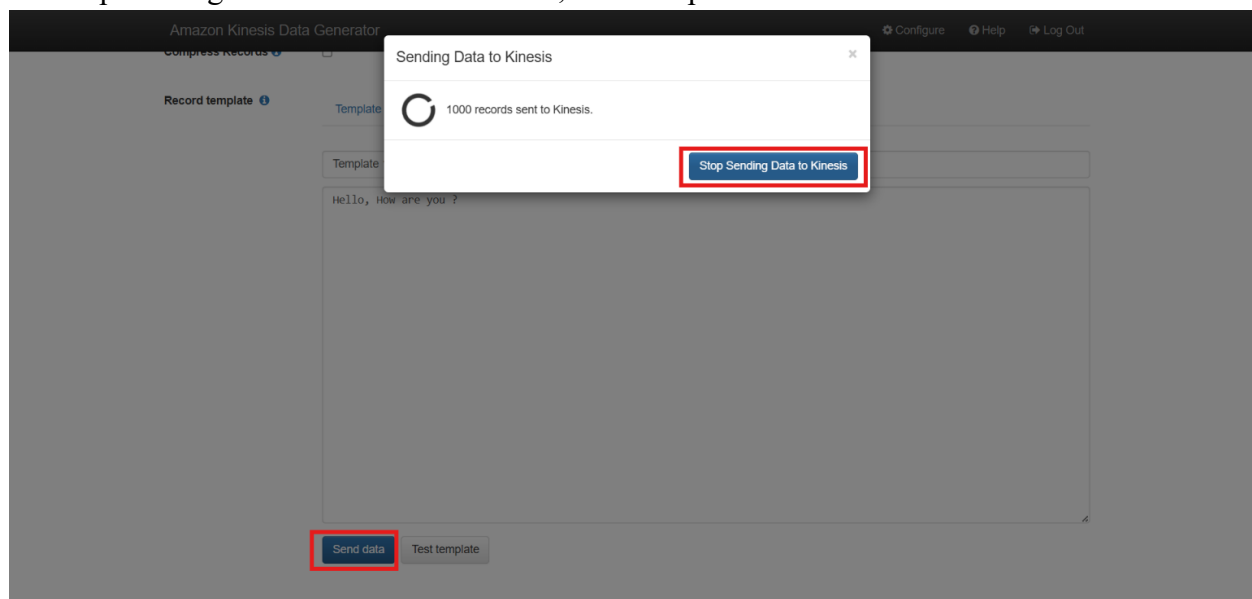
100

Compress Records: ☐

Record template: Template 1 Template 2 Template 3 Template 4 Template 5

Template 1: Hello, How are you ? |

- Click **Send data**.
- Stop sending data after a certain record, I will stop it after 1000 records.



Step 5 – Verify Data in the Stream

- Open the Kinesis stream.
- Go to the **Data viewer**.
- Select a shard.
- Click **Get records**.

Data stream summary

Status: Active

Capacity mode: On-demand

Data retention period: 1 day

Maximum record size: 1024 KiB

ARN: XXXXXXXXXXXXstream/CloudFreeks

Creation time: July 04, 2026 at 11:56 GMT+5:30

Applications | Monitoring | Configuration | Enhanced fan-out (0) | **Data viewer** | Data analytics - new | Data stream sharing | EventBridge Pipes

Shard: shardId-000000000000

Starting position: Trim horizon

[Get records](#)

Records (50)

Shard: shardId-000000000000 Starting position: Trim horizon

Find records

Partition key	Data	Approximate arrival timestamp	Sequence number
7048775550	Hello, How are you ?	July 04, 2026 at 12:11:01 GMT+5:30	4967610520078637081399412377696445857097...
33006561	Hello, How are you ?	July 04, 2026 at 12:11:01 GMT+5:30	4967610520078637081399412377696566749679...
3486574384	Hello, How are you ?	July 04, 2026 at 12:11:01 GMT+5:30	4967610520078637081399412377696687642261...
2009448046	Hello, How are you ?	July 04, 2026 at 12:11:01 GMT+5:30	4967610520078637081399412377696808534843...

- If no records appear immediately, try changing **Starting Position** to **Trim Horizon**, even if the records don't show then try it with another shard.
- Confirm that the record eventually appears in the stream.

Step 6 – Create a Second Stream if Needed

- If the first stream is not showing the expected test behavior, create another stream.
- Enter a new name such as: live stream
- Keep the same on-demand settings.
- Open the producer app again.
- Select the new stream.
- Send data again.
- Check the data viewer to confirm that records are visible.

Step 7 – Clean Up the Lab Resources

- Delete the CloudFormation stack that created the producer app.
- Delete the Kinesis stream if you no longer need it.
- Remove the test stream instances to avoid leaving unused resources behind.

Verification

- A **Kinesis Data Stream** was created successfully.
- The stream was configured in **On-demand** mode.
- A producer app was created with CloudFormation and Cognito.
- Test records were sent into the stream.
- The **Data viewer** showed stream records.
- The lab confirmed that Kinesis receives streaming data and stores it temporarily for consumers to read.